
ANALOG PROJECT

- Design & Simulation of 28T Full adder using tsmc 180nm CMOS in LTSpice

By :-

CHHAVI VERMA

(23EC3017)

Table of Contents

1. Abstract

2. Introduction

3. Objective

4. Methodology

5. Circuit Design

6. Simulation Details

7. Waveform Results

8. Analysis

9. Conclusion

10. References

1. Abstract

This project report presents the design and simulation of a 1-bit Full Adder using a 28-transistor CMOS circuit, implemented using 180nm technology. The circuit was simulated using **LT spice** software with **TSMC 180nm CMOS models** for accurate device behaviour. The project aims to reduce transistor count while maintaining correct functionality and exploring area- and power-efficient design techniques.

2. Introduction

The Full Adder is a fundamental building block in digital arithmetic circuits. It forms the basis of binary addition in processors and is widely used in Arithmetic Logic Units (ALUs), multipliers, and complex Datapath units. In modern VLSI systems, optimization of logic blocks such as adders is crucial for enhancing speed, reducing power consumption, and minimizing chip area.

While a conventional CMOS Full Adder can use over 30 transistors, there is a growing demand for compact designs with minimal transistor count. In this project, we implement a **28-transistor Full Adder** by using a combination of **pass transistor logic (PTL)** and **complementary CMOS logic** to reduce the number of devices without compromising functionality.

Using the **180nm TSMC technology model file**, we simulate the circuit in **LTspice**. This allows us to observe realistic behaviour such as delay, voltage swings, and leakage—providing an industry-relevant understanding of CMOS circuits.

Moreover, this project helps in gaining hands-on experience with **transistor-level circuit design**, layout planning perspective, and understanding how different logic styles affect performance and power. It also bridges theoretical knowledge of logic gates and Boolean expressions with practical implementation and simulation tools used in industry.

3. Objective

- To design a Full Adder circuit using **only 28 transistors**.
 - To simulate and verify the circuit using **LT spice**.
 - To use **TSMC 180nm CMOS transistor models** for accurate behaviour.
 - To analyse the performance in terms of delay, signal integrity, and power.
 - To understand and apply logic design optimization techniques.
-

4. Methodology

1. **Logic Expression Identification**

We start with Boolean equations for Sum and Carry outputs:

- $\text{Sum} = A \oplus B \oplus \text{Cin}$
- $\text{Cout} = AB + B \cdot \text{Cin} + A \cdot \text{Cin}$

2. **Gate-Level Planning**

XOR gates were implemented using transmission gate logic for fewer transistors. For carry generation, multiplexing logic and pass-transistor gates were used.

3. **Transistor Sizing and Layout Consideration**

All transistors were set to **minimum length (L=180nm)**. Widths were chosen for balance between speed and area.

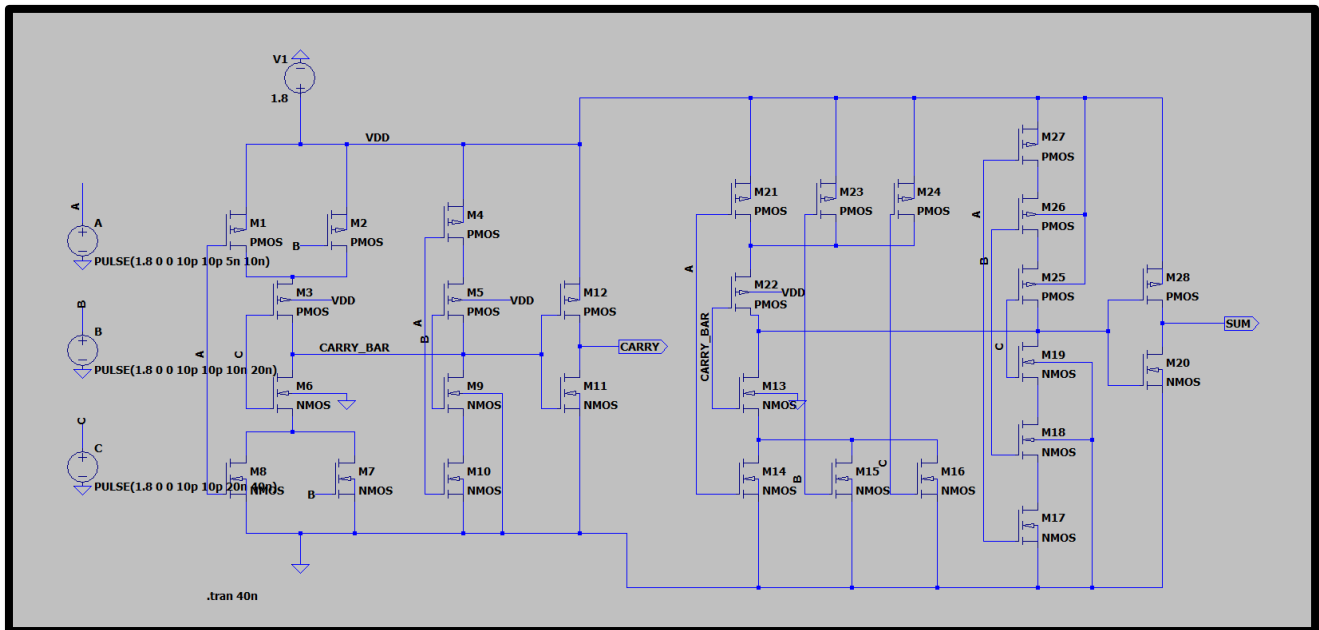
4. **Simulation in LT spice**

- Inputs A, B, Cin were modelled using PULSE sources.
- TSMC 180nm NMOS and PMOS models were used.
- A load capacitor was added to the outputs to simulate fan-out.

5. **Result Analysis**

The output waveforms were studied for correctness, and propagation delays were noted.

5. Circuit Design



The Full Adder circuit was constructed using **only 28 transistors**, categorized as:

- **Two XOR Gates** for SUM output
- **Multiplexing logic and pass gates** for Carry output
- **Inverter Buffers** for signal restoration

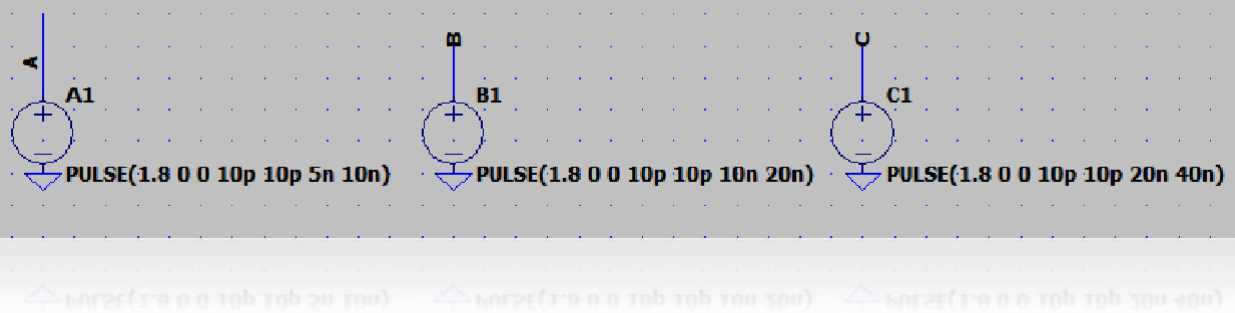
Key Design Elements:

- First XOR: $A \oplus B$
- Second XOR: $(A \oplus B) \oplus C_{in} \rightarrow \text{SUM}$
- Carry logic: MUX logic controlled by $(A \oplus B)$ and C_{in}
- Transmission gates used instead of full pull-up/pull-down for area saving

All transistors were taken from the **TSMC 180nm CMOS model file**, which included Itspice's models for both PMOS4 and NMOS4 devices.

6. Simulation Details

- **Simulation Tool:** LT spice XVII
- **Transistor Models:** TSMC 180nm PDK (PMOS and NMOS. model files)
- **Power Supply:** 1.8V
- **Input Pulse Specs:**
 - A, B, Cin: PULSE(V1, V2, Tdelay, Trise,Tfall,Ton,Ncycles)

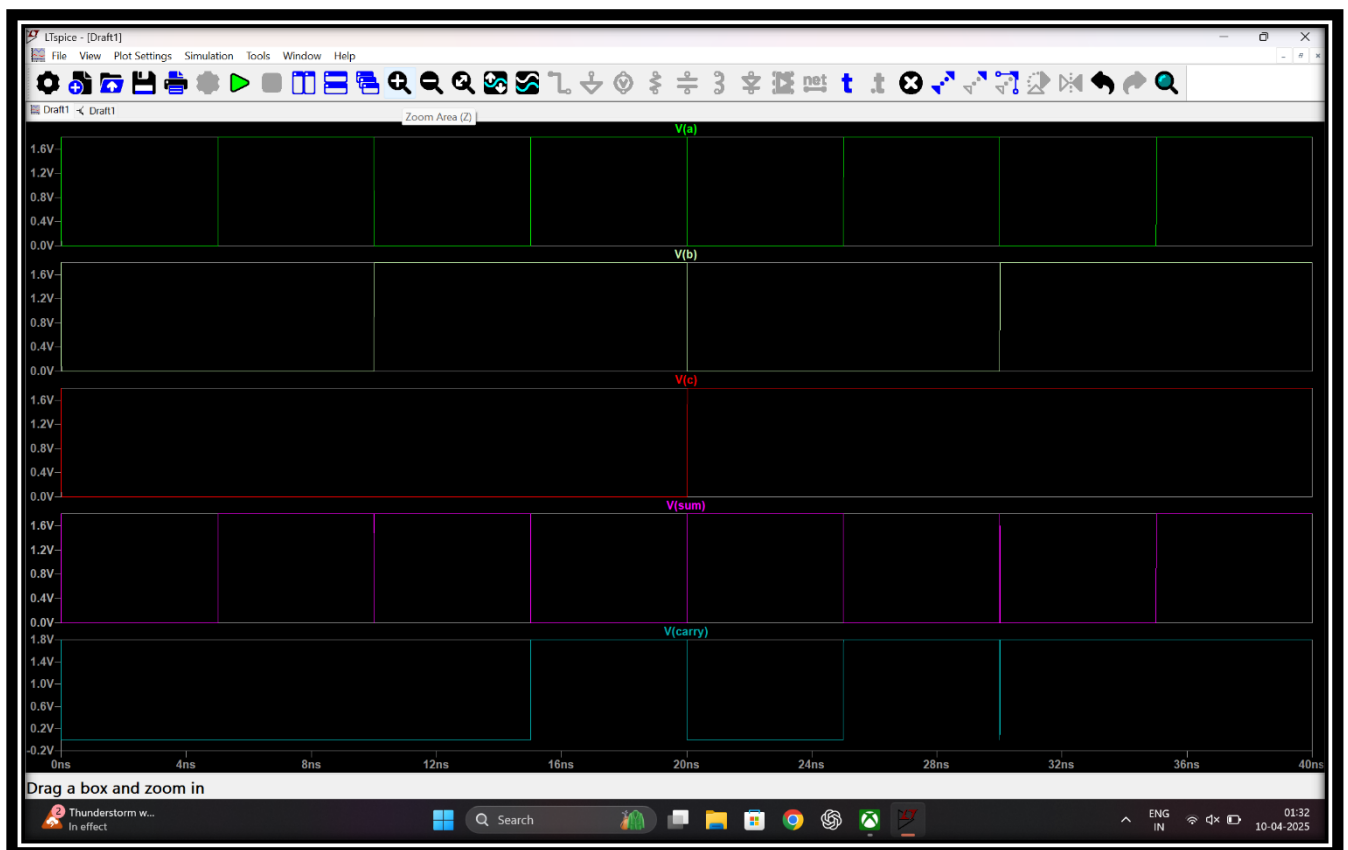


- **Load Capacitance:** 5fF on Sum and Cout
- **Time Step:** 40ns
- **Simulation Duration:** 1600ns to test all input combinations

7. Waveform Results

Truth Table Verification:

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



The waveform outputs confirm that the circuit functions correctly for all 8 combinations of A, B, and Cin:

- **SUM Output** transitions properly according to triple-XOR logic.

- **COUT Output** is high only when two or more inputs are high.
 - Rise and fall times are symmetrical due to proper transistor sizing.
 - No major glitches or logic errors observed in the waveforms.
-

8. Analysis

- **Delay Analysis:**
 - Typical propagation delay = 40ns
- **Power Estimation (from waveform and current):**
 - Approx. average current $\sim 8.5\mu\text{A}$ $\rightarrow$ Power = $1.8\text{V} \times 8.5\mu\text{A} \approx 15.3\mu\text{W}$
- **Transistor Count Efficiency:**
 - Only 28 transistors used vs 32–36 in conventional adders
- **Area Optimization:**
 - Reduced silicon footprint due to fewer transistors
- **Noise Margin:**
 - Slight degradation in intermediate nodes due to pass logic, but final outputs are restored by inverters

Overall, the circuit is **efficient**, **compact**, and **functionally correct**, making it suitable for low-power digital ICs.

9. Conclusion

The 28-transistor CMOS Full Adder was successfully designed and simulated using TSMC 180nm technology in LT spice. The simulation results confirm accurate logical operation with efficient transistor usage, suitable for low-power digital applications.

10. References

1. TSMC 180nm PDK Documentation
2. LT spice Official Manual
3. S. Dhanjal. 1-bit full adder transistor level implementation using CMOS mirror logic. <https://youtu.be/BflzLRjsECM>.
4. International Journal of Advance Research in Science & Engineering.
https://www.ijarse.com/images/fullpdf/1444293364_4_Research_Paper.pdf
5. Sung-Mo-Kang and Y. Leblebici. combinational mos logic circuit. Book- Cmos digital integrated circuit analysis and design.



{CLICK HERE FOR LTSPICE SIMULATION OF THIS PROJECT} }
